

MAKINDU, Kenya

WMO: 637660

Lat: 2.2841S

Lon: 37.8208E

Elev: 3281

StdP: 13.04

Time Zone: 3.00 (E03)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	56.6	58.3	47.7	55.2	77.3	49.5	59.1	77.7	18.3	75.9	15.4	73.8	2.8	270	0.259

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	20.4	90.2	66.9	88.9	67.1	87.5	67.2	71.7	81.1	71.0	80.2	70.6	79.5	7.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.5	122.7	73.5	68.6	118.8	72.7	68.3	117.4	72.5	37.7	81.2	37.2	80.5	36.7	79.9	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	12.8	10.9	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.8	74.5	76.9	77.4	76.1	73.6	71.1	69.8	70.3	72.9	75.0	74.0	74.1		
	DBStd	3.18	2.21	2.11	2.11	2.09	1.69	1.95	1.98	2.45	2.34	2.12	2.51	2.20		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	3	0	0	0	0	0	0	0	3	0	0	0	0		
	CDD50	8674	758	752	848	782	731	632	614	630	687	774	719	747		
	CDD65	3201	293	332	383	332	266	182	149	167	237	309	269	282		
	CDH74	21666	2040	2854	3198	2228	1644	1096	783	1066	1558	2394	1448	1357		
	CDH80	5901	591	1136	1229	622	327	143	63	160	354	756	308	212		
(14) Wind	WSAvg	5.0	4.7	4.5	4.9	5.0	4.2	4.7	4.9	5.4	6.3	6.2	4.7	4.1		
(15) Precipitation	PrecAvg	22.9	1.5	1.1	2.4	4.3	0.9	0.1	0.0	0.0	0.1	1.2	7.1	4.3		
	PrecMax	47.6	7.5	5.3	8.1	11.3	3.9	0.8	0.2	0.7	0.6	8.7	18.8	12.5		
	PrecMin	10.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.1		
	PrecStd	9.7	1.7	1.3	2.3	3.0	0.9	0.2	0.0	0.1	0.2	1.9	4.6	3.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.0	92.4	91.6	89.7	86.3	84.9	84.2	85.9	87.5	89.4	87.9	85.9		
		MCWB	65.7	67.2	67.2	68.0	68.4	66.0	63.2	63.8	64.4	66.8	67.6	69.1		
	2%	DB	88.1	90.6	89.9	87.3	84.6	82.8	82.1	83.4	85.6	87.6	85.8	84.1		
		MCWB	66.4	66.7	67.3	68.6	67.9	65.1	63.5	63.6	64.1	66.3	67.5	69.4		
	5%	DB	86.0	88.9	88.4	85.6	83.5	81.5	80.4	81.7	84.0	85.9	83.9	82.4		
		MCWB	66.4	67.0	67.7	68.6	67.6	64.9	63.1	63.3	64.1	65.7	67.7	69.4		
	10%	DB	83.6	87.0	86.7	84.0	82.1	80.2	78.6	80.0	82.2	84.2	81.7	80.8		
		MCWB	67.2	67.3	67.9	68.6	67.4	64.7	62.6	63.1	63.9	65.4	68.0	69.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.8	71.9	72.3	72.2	71.8	70.0	67.1	68.5	68.3	70.1	71.5	71.9		
		MCDB	80.2	83.7	82.6	81.4	81.0	79.9	78.8	79.4	80.3	80.6	80.1	80.5		
	2%	WB	70.6	70.9	71.1	71.1	70.6	67.7	65.1	66.1	66.5	68.7	70.6	71.1		
		MCDB	79.0	82.2	81.1	80.3	79.8	77.6	76.5	77.0	78.8	78.6	78.7	79.3		
	5%	WB	69.7	70.0	70.5	70.4	69.8	66.6	64.3	64.8	65.7	67.9	69.9	70.5		
		MCDB	78.3	81.1	80.1	79.5	78.8	76.1	75.3	76.3	77.5	77.6	77.6	78.4		
	10%	WB	69.0	69.2	69.8	69.8	69.0	65.8	63.7	63.9	64.9	67.2	69.2	70.0		
		MCDB	77.6	79.9	79.1	78.7	77.6	75.1	74.3	75.4	76.4	77.2	76.5	77.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.5	21.6	20.4	17.8	18.0	19.3	19.4	20.7	20.3	20.7	15.7	14.6		
		MCDBR	23.9	24.6	22.7	19.7	20.3	22.2	21.9	23.6	22.5	22.7	19.4	16.8		
		MCWBR	6.9	6.1	5.6	4.7	5.9	7.0	6.3	6.8	5.4	5.3	4.8	4.6		
	5% WB	MCDBR	17.5	20.6	19.5	17.6	17.2	18.9	19.0	21.4	20.4	20.0	15.1	14.9		
		MCWBR	6.9	6.0	5.7	5.2	6.2	6.7	6.2	7.6	5.7	5.5	5.1	5.2		
(40) Clear-Sky Solar Irradiance	taub		0.399	0.419	0.398	0.349	0.343	0.376	0.388	0.373	0.360	0.364	0.371	0.393		
	taud		2.402	2.327	2.416	2.599	2.612	2.463	2.392	2.424	2.461	2.472	2.493	2.441		
	Ebn at Noon		296	290	294	301	295	280	279	290	300	303	303	297		
	Edh at Noon		39	43	39	31	30	34	37	37	37	37	36	38		
(44) All-Sky Solar Radiation	RadAvg		1986	2070	2021	1814	1623	1460	1449	1551	1846	1939	1835	1873		
	RadStd		110	96	120	82	60	84	120	111	108	94	75	92		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.85	+1.24	N/A	+1.03	+0.70	+0.29	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page